

Exponentials & Logarithms



Evaluating exponential expressions

- 1 Evaluate 2^5 .
 - 2 Evaluate 3^{-2} .
 - 3 Evaluate $16^{1/2}$.
 - 4 Evaluate $27^{2/3}$.
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Evaluating logarithms

- 5 Evaluate $\log_2 32$.
 - 6 Evaluate $\log_3 81$.
 - 7 Evaluate $\log_5 1$.
 - 8 Evaluate $\log_{10} 0.01$.
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Log laws: product and quotient

- 9 Write $\log_2 (8x)$ as a sum of logarithms.
 - 10 Write $\log_3 \left(\frac{x}{9}\right)$ as a difference of logarithms.
 - 11 Combine $\log_5 5 + \log_5 25$ into a single logarithm and evaluate.
 - 12 Combine $\log_2 24 - \log_2 3$ into a single logarithm and evaluate.
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Log laws: the power rule

- 13 Write $\log(x^5)$ using the power rule.
- 14 Evaluate $\log_2 (4^3)$ using the power rule.
- 15 Write $2\log_3 x + \log_3 y$ as a single logarithm.

Expanding and condensing logarithmic expressions

16 Expand $\log\left(\frac{x^3y}{z^2}\right)$ fully.

17 Condense $3\ln x - 2\ln y$ into a single logarithm.

18 Expand $\log_2(\sqrt{x} \cdot y^3)$ fully.

Solving exponential equations (matching bases)

19 Solve $2^x = 32$.

20 Solve $3^{x+1} = 81$.

21 Solve $5^{2x} = 125$.

Solving exponential equations using logarithms

22 Solve $2^x = 10$ for x , rounded to three decimal places.

23 Solve $5^x = 40$ for x , rounded to three decimal places.

24 Solve $3^{2x} = 50$ for x , rounded to three decimal places.

25 Solve $e^x = 15$ for x , rounded to three decimal places.

Solving logarithmic equations

26 Solve $\log_2(x) = 5$.

27 Solve $\log_3(x - 1) = 2$.

28 Solve $\log(x) + \log(x - 3) = 1$.

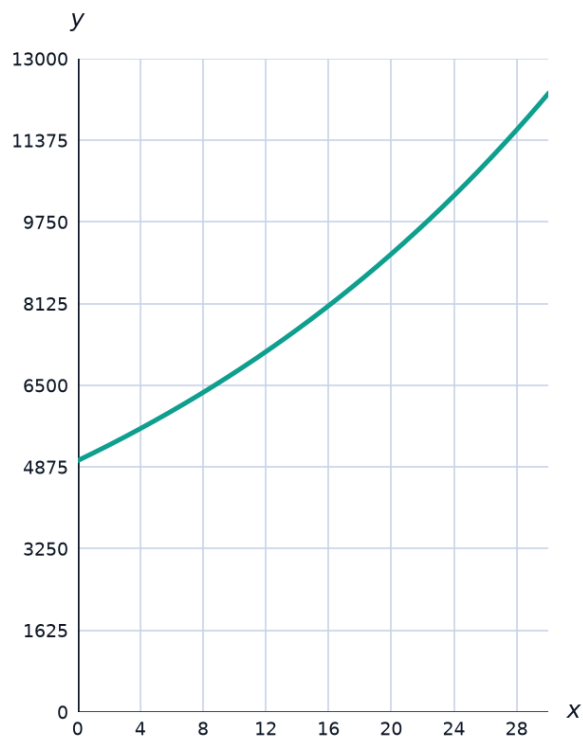
29 Solve $\log_2(x + 3) + \log_2(x - 1) = 5$.

Natural logarithms and e

- 30 Evaluate $\ln(e^3)$.
- 31 Evaluate $\ln(1)$.
- 32 Solve $\ln(x) = 2$ for x , in terms of e .
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Exponential growth and decay models

- 33 The population of a town is modeled by $P(t) = 5000e^{0.03t}$, where t is in years. Find $P(10)$, rounded to the nearest whole number.



- 34 A radioactive substance decays according to $A(t) = 200e^{-0.05t}$ (grams). Find $A(20)$, rounded to one decimal place.
- 35 Determine whether $N(t) = 50e^{0.02t}$ models growth or decay, and justify your answer.
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Word problems: exponential models in context

- 36** This question has two parts. An investment of 2000 dollars grows according to $A(t) = 2000e^{0.04t}$, where t is in years.
- a** Find the value of the investment after 5 years, rounded to the nearest dollar.
 - b** Find how many years it takes for the investment to double, rounded to one decimal place (using $\ln 2 \approx 0.693$).
- 37** This question has two parts. A cup of coffee cools according to $T(t) = 20 + 60e^{-0.1t}$ ($^{\circ}\text{C}$), where t is minutes after brewing and 20°C is room temperature.
- a** Find the temperature after 15 minutes, rounded to one decimal place (using $e^{-1.5} \approx 0.2231$).
 - b** Find how many minutes it takes for the temperature to drop to 35°C , rounded to one decimal place (using $\ln 4 \approx 1.386$).
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MCQ — Evaluating exponential expressions

- 38** Evaluate 3^4 .
- a.** 81
 - b.** 243
 - c.** 12
 - d.** 64
- 39** Evaluate 4^{-2} .
- a.** $-\frac{1}{16}$
 - b.** $\frac{1}{16}$
 - c.** $\frac{1}{8}$
 - d.** 16

40 Evaluate $25^{1/2}$.

- a. 5
- b. 50
- c. 12.5
- d. 625

41 Evaluate $8^{2/3}$.

- a. 5.33
- b. 4
- c. 2
- d. 16

42 Evaluate 2^{-3} .

- a. -8
- b. $\frac{1}{6}$
- c. 8
- d. $\frac{1}{8}$

43 Evaluate $32^{3/5}$.

- a. 2
- b. 16
- c. 8
- d. 32

MCQ — Evaluating logarithms

44 Evaluate $\log_3 27$.

a. 3

b. 1

c. 27

d. 9

45 Evaluate $\log_2 64$.

a. 32

b. 8

c. 5

d. 6

46 Evaluate $\log_7 1$.

a. 1

b. Undefined

c. 7

d. 0

47 Evaluate $\log_{10} 0.001$.

a. 0.001

b. -3

c. -2

d. 3

48 Evaluate $\log_4 16$.

- a. 4
- b. 8
- c. 2
- d. 16

49 Evaluate $\log_5 125$.

- a. 5
- b. 3
- c. 1
- d. 25

MCQ — Log laws: product and quotient

50 Write $\log_3 (9x)$ as a sum of logarithms.

- a. $2 + \log_3 x$
- b. $2\log_3 x$
- c. $9 + \log_3 x$
- d. $\log_3 9 + \log_3 x$ only

51 Write $\log_5 \left(\frac{x}{25} \right)$ as a difference of logarithms.

- a. $\log_5 x - 25$
- b. $2 - \log_5 x$
- c. $\log_5 x - 2$
- d. $\frac{\log_5 x}{2}$

52 Combine $\log_2 4 + \log_2 8$ into a single logarithm and evaluate.

- a. 32
- b. 5
- c. 12
- d. 2

53 Combine $\log_3 54 - \log_3 2$ into a single logarithm and evaluate.

- a. 27
- b. 3
- c. 26
- d. 52

MCQ — Log laws: the power rule

54 Write $\log(x^7)$ using the power rule.

- a. $7\log x$
- b. $x^7\log x$
- c. $\log x^7$ unchanged
- d. $\log(7x)$

55 Evaluate $\log_3 (9^2)$ using the power rule.

- a. 18
- b. 81
- c. 2
- d. 4

56 Write $3\log_2 x + \log_2 y$ as a single logarithm.

- a. $3\log_2(xy)$
- b. $\log_2(x^3 + y)$
- c. $\log_2(x^3y)$
- d. $\log_2(3xy)$

57 Write $4\log x - \log y$ as a single logarithm.

- a. $\log\left(\frac{x^4}{y}\right)$
- b. $\log(4x - y)$
- c. $4\log\left(\frac{x}{y}\right)$
- d. $\log\left(\frac{4x}{y}\right)$

MCQ — Expanding and condensing logarithmic expressions

58 Expand $\log\left(\frac{x^2y}{z^3}\right)$ fully.

- a. $2\log x + \log y - 3\log z$
- b. $2\log x - \log y - 3\log z$
- c. $\log x^2 + \log y - \log z^3$ unchanged
- d. $2\log x + \log y + 3\log z$

59 Condense $2\ln x - 3\ln y$ into a single logarithm.

- a. $\ln\left(\frac{x^2}{y^3}\right)$
- b. $\ln\left(\frac{x^2}{y}\right) - 3$
- c. $\ln\left(\frac{x^3}{y^2}\right)$
- d. $\ln(x^2 - y^3)$

60 Expand $\log_2 (\sqrt{x} \cdot y^4)$ fully.

- a. $\frac{1}{2}\log_2 x + 4\log_2 y$
- b. $\frac{1}{2}\log_2 x + \log_2 y^4$ unchanged
- c. $2\log_2 x + 4\log_2 y$
- d. $\frac{1}{4}\log_2 x + 2\log_2 y$

61 Condense $\frac{1}{2}\log x + 2\log y$ into a single logarithm.

- a. $\log(x \cdot y^2)$
- b. $\log\left(\frac{\sqrt{x}}{y^2}\right)$
- c. $\log(\sqrt{x} \cdot y^2)$
- d. $\log(2\sqrt{x} \cdot y)$

MCQ — Solving exponential equations (matching bases)

62 Solve $2^x = 64$.

- a. $x = 8$
- b. $x = 6$
- c. $x = 32$
- d. $x = 5$

63 Solve $3^{x+2} = 27$.

- a. $x = 9$
- b. $x = 1$
- c. $x = 5$
- d. $x = 3$

64 Solve $5^{3x} = 125$.

a. $x = \frac{1}{3}$

b. $x = 41.67$

c. $x = 3$

d. $x = 1$

65 Solve $4^x = \frac{1}{16}$.

a. $x = 4$

b. $x = -4$

c. $x = -2$

d. $x = 2$

66 Solve $2^{2x-1} = 32$.

a. $x = 3$

b. $x = 2.5$

c. $x = 6$

d. $x = 5.5$

MCQ — Solving exponential equations using logarithms

67 Solve $2^x = 12$ for x , rounded to three decimal places.

a. 3.585

b. 3.170

c. 4.000

d. 2.485

68 Solve $7^x = 50$ for x , rounded to three decimal places.

- a. 7.143
- b. 2.010
- c. 3.000
- d. 1.505

69 Solve $4^{2x} = 90$ for x , rounded to three decimal places.

- a. 6.492
- b. 3.246
- c. 1.623
- d. 0.811

70 Solve $e^x = 25$ for x , rounded to three decimal places.

- a. 25.000
- b. 2.303
- c. 3.219
- d. 12.500

MCQ — Solving logarithmic equations

71 Solve $\log_3(x) = 4$.

- a. $x = 7$
- b. $x = 12$
- c. $x = 64$
- d. $x = 81$

72 Solve $\log_2(x - 1) = 3$.

a. $x = 6$

b. $x = 9$

c. $x = 7$

d. $x = 8$

73 Solve $\log(x) + \log(x - 3) = 1$.

a. $x = -2$

b. $x = 5$

c. $x = 10$

d. $x = 13$

74 Solve $\log_2(x + 2) + \log_2(x - 2) = 5$.

a. $x = 6$

b. $x = -6$

c. $x = 34$

d. $x = 18$

MCQ — Natural logarithms and e

75 Evaluate $\ln(e^5)$.

a. 0

b. e^5

c. 1

d. 5

76 Evaluate $\ln\left(\frac{1}{e}\right)$.

- a. 1
- b. e
- c. -1
- d. 0

77 Solve $\ln(x) = 3$ for x , in terms of e .

- a. $x = e^{1/3}$
- b. $x = 3$
- c. $x = e^3$
- d. $x = 3e$

78 Evaluate $e^{\ln 7}$.

- a. 7
- b. $\ln 7$
- c. 1
- d. e^7

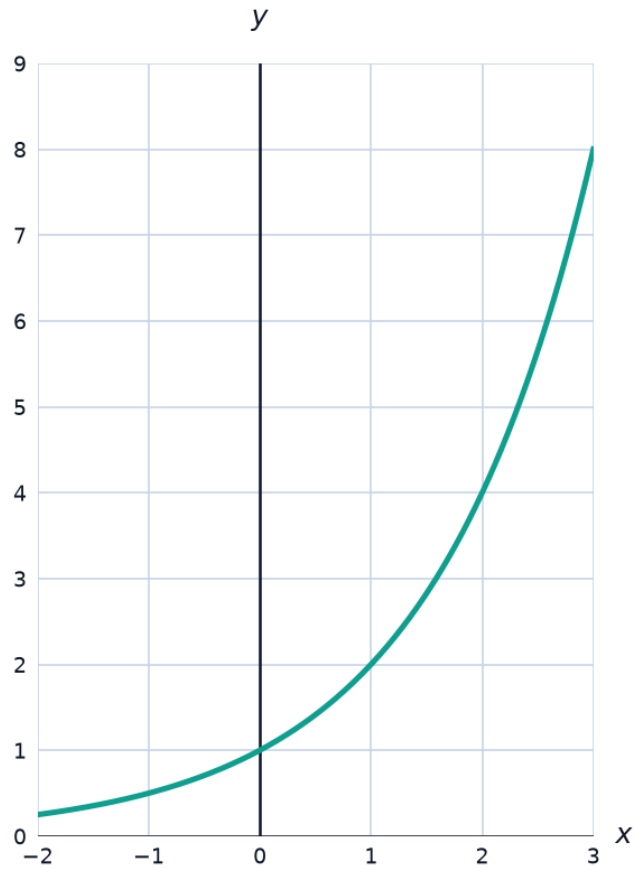
MCQ — Exponential growth and decay models

79 The population of a town is modeled by $P(t) = 8,000e^{0.025t}$, where t is in years. Find $P(15)$, rounded to the nearest whole number.

- a. 8,300
- b. 11,000
- c. 11,640
- d. 12,000

- 80 A radioactive substance decays according to $A(t) = 150e^{-0.04t}$ (grams). Find $A(25)$, rounded to one decimal place.
- a. 60.7
 - b. 50.0
 - c. 75.0
 - d. 55.2
- 81 Determine whether $N(t) = 80e^{-0.015t}$ models growth or decay.
- a. Decay
 - b. Neither
 - c. Growth
 - d. Cannot be determined
- 82 Determine whether $M(t) = 30e^{0.01t}$ models growth or decay.
- a. Neither
 - b. Cannot be determined
 - c. Decay
 - d. Growth

83 The graph shows an exponential function. Based on the graph, does it represent growth or decay?



- a. Neither
- b. Growth
- c. Cannot be determined
- d. Decay

MCQ — Word problems: exponential models in context

- 84 An investment of 3,000 dollars grows according to $A(t) = 3,000e^{0.05t}$ dollars, where t is in years. Find the investment's value after 8 years, rounded to the nearest dollar.
- a. 4,200
 - b. 4,800
 - c. 4,475
 - d. 3,400
- 85 A cup of coffee cools according to $T(t) = 22 + 70e^{-0.08t}$ ($^{\circ}\text{C}$), where t is in minutes. Find the temperature after 15 minutes, rounded to one decimal place.
- a. 38.5°C
 - b. 50.0°C
 - c. 43.1°C
 - d. 22.0°C
- 86 A population of bacteria doubles every 3 hours, starting at 500. Using $P(t) = 500 \cdot 2^{t/3}$, find the population after 9 hours.
- a. 4,000
 - b. 2,000
 - c. 8,000
 - d. 1,500
- 87 The value of a car depreciates according to $V(t) = 25,000(0.85)^t$ dollars, where t is in years. Find the car's value after 5 years, rounded to the nearest dollar.
- a. 11,093
 - b. 12,750
 - c. 9,500
 - d. 10,625

Multiple choice — easy

88 Evaluate 5^{-3} .

- a. $\frac{1}{125}$
- b. $-\frac{1}{125}$
- c. 125
- d. $\frac{1}{127}$

89 Evaluate $4^{\frac{1}{2}}$.

- a. 4
- b. 5
- c. 2
- d. 6

90 Evaluate 5^{-2} .

- a. 25
- b. $\frac{1}{27}$
- c. $-\frac{1}{25}$
- d. $\frac{1}{25}$

91 Evaluate 5^2 .

- a. 27
- b. 10
- c. 23
- d. 25

92 Evaluate 3^{-2} .

a. $\frac{1}{11}$

b. $-\frac{1}{9}$

c. $\frac{1}{9}$

d. 9

93 Evaluate $\log_2 4$.

a. 3

b. 1

c. 2

d. 4

94 Evaluate $\log_4 1$.

a. 0

b. 4

c. -1

d. 1

95 Evaluate $\log_{10} 0.01$.

a. 2

b. -2

c. -3

d. -1

96 Evaluate $\log_{10} 0.1$.

a. -2

b. -1

c. -3

d. 1

97 Evaluate $\log_5 625$.

a. 625

b. 3

c. 5

d. 4

98 Solve $4^x = 256$.

a. 4

b. 3

c. 6

d. 5

99 Solve $3^{x+3} = 81$.

a. 4

b. 1

c. 2

d. 0

100 Solve $4^{x+3} = 64$.

- a. 1
- b. -1
- c. 0
- d. 3

101 Solve $5^x = 125$.

- a. 5
- b. 3
- c. 2
- d. 4

102 Solve $4^{x-3} = 16$.

- a. 5
- b. 6
- c. 2
- d. 4

Multiple choice — medium

103 Combine $\log_2 2 + \log_2 2$ into a single logarithm and evaluate.

- a. 3
- b. 1
- c. 4
- d. 2

104 Combine $\log_5 125 + \log_5 125$ into a single logarithm and evaluate.

- a. 7
- b. 5
- c. 15625
- d. 6

105 Combine $\log_5 125 - \log_5 25$ into a single logarithm and evaluate.

- a. 3
- b. 5
- c. 1
- d. 2

106 Combine $\log_5 5 + \log_5 25$ into a single logarithm and evaluate.

- a. 125
- b. 2
- c. 3
- d. 4

107 Combine $\log_5 125 + \log_5 25$ into a single logarithm and evaluate.

- a. 5
- b. 6
- c. 3125
- d. 4

108 Evaluate $\log_2 (8^2)$ using the power rule.

- a. 6
- b. 8
- c. 5
- d. 4

109 Evaluate $\log_2 (4^4)$ using the power rule.

- a. 6
- b. 8
- c. 12
- d. 4

110 Evaluate $\log_3 (9^4)$ using the power rule.

- a. 4
- b. 8
- c. 12
- d. 6

111 Evaluate $\log_3 (27^3)$ using the power rule.

- a. 12
- b. 9
- c. 11
- d. 6

112 Evaluate $\log_3 (27^2)$ using the power rule.

- a. 5
- b. 4
- c. 8
- d. 6

113 Solve $\log_5 (x - 2) = 4$.

- a. 629
- b. 630
- c. 627
- d. 625

114 Solve $\log_4 (x - 1) = 4$.

- a. 257
- b. 255
- c. 256
- d. 259

115 Solve $\log_5 (x) = 3$.

- a. 123
- b. 128
- c. 125
- d. 127

116 Solve $\log_2(x - 1) = 3$.

- a. 9
- b. 8
- c. 7
- d. 11

117 Solve $\log_4(x - 2) = 2$.

- a. 21
- b. 18
- c. 16
- d. 20

Multiple choice — hard

118 Solve $3^x = 80$ for x , rounded to three decimal places.

- a. 4.090
- b. 3.890
- c. 3.990
- d. 1.903

119 Solve $2^x = 20$ for x , rounded to three decimal places.

- a. 4.222
- b. 4.422
- c. 4.322
- d. 1.301

120 Solve $5^x = 50$ for x , rounded to three decimal places.

- a. 2.531
- b. 2.331
- c. 2.431
- d. 1.699

121 Solve $3^x = 50$ for x , rounded to three decimal places.

- a. 3.662
- b. 1.699
- c. 3.562
- d. 3.462

122 Solve $2^x = 70$ for x , rounded to three decimal places.

- a. 1.845
- b. 6.130
- c. 6.030
- d. 6.230

123 Solve $\ln(x) = 3$ for x , in terms of e .

- a. e^3
- b. $3e$
- c. e^4
- d. e^2

124 Evaluate $\ln(e^6)$.

- a. 5
- b. 7
- c. 6
- d. e^6

125 Evaluate $\ln(e^4)$.

- a. 3
- b. 5
- c. 4
- d. e^4

126 Evaluate $\ln(e^3)$.

- a. 2
- b. 3
- c. 4
- d. e^3

127 Evaluate $\ln(e^2)$.

- a. 1
- b. e^2
- c. 2
- d. 3

- 128** A quantity is modeled by $N(t) = 500e^{-0.1t}$. Find $N(5)$, rounded to one decimal place.
- a. 273.0
 - b. 500
 - c. 303.3
 - d. 333.6
- 129** A quantity is modeled by $N(t) = 200e^{0.1t}$. Find $N(5)$, rounded to one decimal place.
- a. 296.7
 - b. 200
 - c. 362.7
 - d. 329.7
- 130** A quantity is modeled by $N(t) = 5000e^{0.1t}$. Find $N(5)$, rounded to the nearest whole number.
- a. 8244
 - b. 5000
 - c. 9068.4
 - d. 7419.6
- 131** A quantity is modeled by $N(t) = 1000e^{-0.1t}$. Find $N(5)$, rounded to one decimal place.
- a. 545.9
 - b. 606.5
 - c. 667.1
 - d. 1000

132 A quantity is modeled by $N(t) = 500e^{0.1t}$. Find $N(5)$, rounded to one decimal place.

a. 824.4

b. 742.0

c. 500

d. 906.8